Cover sheet for submission of work for assessment

SWINBURNE UNIVERSITY OF TECHNOLOGY

UNIT DETAILS

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Table of Contents

Table of Contents 2
1. Instructions 3
1.1 System Requirements 3
1.2 Setup and Launch 3
1.3 GUI Overview - Five Tabs 3
2. Introduction 3
3. Features/Bugs/Missing 5
3.1 Implemented Features 5
3.2 Known Bugs and Limitations 6
4. Testing 6
4.1 Test Case Overview 6
4.2 Testcase results 7
4.3 Notable Observations from Testing 7
5. Insights 7
5.1 Dataset and Preprocessing 7
5.2 Model Evaluation Metrics 7
5.3 MAPE Disclaimer 8
5.4 Model Comparison and Discussion 8
5.5 Training Convergence 8
6. Research Initiatives 9
6.1 OpenStreetMap Route Visualisation with folium 9
6.2 Integrated GUI Test Suite 10
6.3 Enhanced GRU Architecture 10
6.4 Custom Graph Construction via Two-Rule Adjacency Inference 10
6.5 Yen's K-Shortest Loopless Paths Algorithm 10
6.6 BiLSTM as an Additional Experimental Model 11
7. Conclusion 11
8. Acknowledgements & Resources: 11
References 12

1. Instructions

1.1 System Requirements

The TBRGS requires Python 3.10 or later. Install all dependencies by running pip install -r requirements.txt from the Assignment2B directory. Key libraries include PyTorch, Streamlit, streamlit-folium, folium, scikit-learn, pandas, numpy, xlrd, and joblib.

1.2 Setup and Launch

Before first use, build the road network graph once by running python preprocessing/graph_builder.py. Then launch the GUI with streamlit run gui/app.py. Models can also be trained via CLI using python training/train.py --model lstm --epochs 100 (replace lstm with gru, bilstm, or transformer). All system defaults are managed through config.ini, editable either directly or from the Settings tab in the GUI.

1.3 GUI Overview - Five Tabs

Route Planner: Enter origin and destination SCATS site number, select a model, and specify date/time. The system returns top five routes ranked by estimated travel time, displayed on an interactive map.

Train Models: Select a model architecture, configure hyperparameters (epochs, batch size, learning rate), and trigger training directly from the browser with live progress output.

Evaluate: View a side-by-side metrics table (MAE, RMSE, MAPE) and predicted-versus-actual charts for all trained models without running any separate script.

Settings: Edit config.ini parameters (lookback window, split sizes, routing defaults) from within the GUI; changes are saved back to the file automatically.

Test Suite: Select any combination of test cases (TC01-TC10) and model architectures and run them as a batch. Structured pass/fail results with full log output are displayed in the browser, no CLI required.

2. Introduction

Traffic congestion is a persistent urban challenge, making accurate short-term traffic forecasting and effective route optimisation increasingly important (Lv et al., 2015). This project proposes a Traffic-Based Route Guidance System that combines deep learning-based traffic flow prediction with graph-based routing to recommend optimal paths between intersections in the Boroondara area of Melbourne.

The system uses the VicRoads SCATS dataset for October 2006, which records traffic volume every 15 minutes across 96 daily intervals (V00-V95). During preprocessing, these records are converted into fixed-length time-series sequences using a sliding window of 12 time steps, split chronologically into training, validation, and test sets, and normalised using MinMaxScaler fitted exclusively on training data to prevent data leakage.

Four deep learning architectures were implemented and compared for traffic flow prediction:

LSTM (Long Short-Term Memory) is a type of recurrent neural network designed to model sequential data by capturing long-term dependencies. It uses memory cells and gating mechanisms (input, forget, and output gates) to retain relevant information and avoid the vanishing gradient problem (Coursera, 2026). In this project, a stacked LSTM is used to learn temporal patterns in traffic flow data and predict the next time-step value based on previous observations.

```
CLASS LSTMRegressor (Inherits BaseTimeSeriesModel):

    FUNCTION Initialize(input_dim, hidden_size, num_layers, dropout):
        Store hyperparameters
        Initialize LSTM layer (stacked layers with dropout)
        Initialize Dropout layer
        Initialize Fully Connected (Linear) layer (hidden_size -> 1)

    FUNCTION ForwardPass(input_x):
        # input_x shape: (batch, sequence_length)
        IF input_x is 2D:
            Add feature dimension -> (batch, sequence_length, 1)

        # Process through LSTM
        LSTM_outputs, _ = LSTM_Layer(input_x)

        # Extract features
        Extract hidden state of the LAST timestep -> (batch, hidden_size)
        Apply Dropout to the extracted state

        # Generate prediction
        Return Linear_Layer(result) -> (batch, 1)

    FUNCTION GetParameters():
        Return dictionary of initialization arguments
```

Figure 1: LSTM pseudocode

A Gated Recurrent Unit (**GRU**) is a streamlined recurrent neural network that captures temporal dependencies in sequential data using fewer parameters than LSTM (Indicio Technologies, n.d.). This makes it computationally efficient and suitable for traffic flow forecasting on smaller datasets. In our implementation, the GRU model uses stacked recurrent layers together with dual temporal pooling, a residual skip path, LayerNorm, and a regression head to predict the next time-step traffic flow value.

```
CLASS GRURegressor (Inherits BaseTimeSeriesModel):

    FUNCTION Initialize(input_dim, hidden_size, num_layers, dropout, head_hidden_dim):
        Store hyperparameters
        Initialize GRU layer (stacked with inter-layer dropout)
        Initialize Residual Projection (Maps H -> 2H)
        Initialize Layer Normalization
        Initialize MLP Head (Sequential: Linear -> GELU -> Dropout -> Linear)
        Reset parameters (Xavier for input, Orthogonal for hidden, Near-zero for Skip)

    FUNCTION ForwardPass(input_x):
        # input_x shape: (batch, sequence_length)
        IF input_x is 2D:
            Add feature dimension -> (B, T, 1)

        # Recurrent backbone
        GRU_outputs, _ = GRU_Layer(input_x) # (B, T, H)

        # Dual Temporal Pooling
        Last_Step_Feature = GRU_outputs[:, -1, :] # Recency bias
        Global_Mean_Feature = Mean(GRU_outputs along time dim) # Global context

        # Feature Fusion & Residual Skip
        Concatenated_Feature = Concat(Last_Step_Feature, Global_Mean_Feature) # (B, 2H)
        Final_Feature = Concatenated_Feature + Residual_Projection(Last_Step_Feature)

        # Final Regression
        Normalized_Feature = LayerNorm(Final_Feature)
        Return MLP_Head(Normalized_Feature) -> (B, 1)

    FUNCTION GetInitKwargs():
        Return dict of params for checkpoint compatibility
```

Figure 2: GRU pseudocode

A Bidirectional LSTM (**BiLSTM**) extends LSTM by learning from both past and future contexts simultaneously (Rahman et al., 2021). In this implementation, a stacked bidirectional LSTM extracts temporal features from traffic data, and the final timestep representation is used to predict the next traffic flow value.

```
CLASS BiLSTMRegressor (Inherits BaseTimeSeriesModel):

    FUNCTION Initialize(input_dim, hidden_size, num_layers, dropout):
        Store hyperparameters
        Initialize BiLSTM layer (bidirectional=True, stacked layers)
        Initialize Dropout layer
        # FC input is 2 * hidden_size due to concatenated forward/backward states
        Initialize Fully Connected (Linear) layer (hidden_size * 2 -> 1)

    FUNCTION ForwardPass(input_x):
        # input_x shape: (batch, sequence_length)
        IF input_x has 2 dimensions:
            Reshape to (batch, sequence_length, 1)

        # BiLSTM processes sequence in both directions
        BiLSTM_outputs, _ = BiLSTM_Layer(input_x) # (B, T, H * 2)

        # Extract features from the final timestep
        Final_Timestep_State = BiLSTM_outputs[:, -1, :] # (B, H * 2)
        Apply Dropout to the extracted state

        # Map fused features to single prediction
        Return Linear_Layer(result) -> (batch, 1)

    FUNCTION GetInitKwargs():
        Return dictionary of constructor arguments for persistence
```

Figure 3: BiLSTM pseudocode

The **Transformer** uses self-attention to model relationships across all time steps simultaneously (Stryker & Bergmann, n.d.). In this implementation, a Transformer encoder extracts temporal features from traffic data, and the final timestep representation is used to predict the next traffic flow value.

```
CLASS TransformerRegressor (Inherits BaseTimeSeriesModel):

    FUNCTION Initialize(input_dim, seq_len, d_model, nhead, num_layers, dropout):
        Store hyperparameters
        Initialize Input Projection (Linear: input_dim -> d_model)
        Initialize Positional Encoding (Learnable parameter: 1 x seq_len x d_model)

        # Setup Transformer Backbone
        Create TransformerEncoderLayer (Multi-head attention + Feedforward + GELU)
        Initialize TransformerEncoder (Stacked layers)

        Initialize Layer Normalization
        Initialize Output Head (Linear: d_model -> 1)

    FUNCTION ForwardPass(input_x):
        # input_x shape: (batch, current_seq_len, input_dim)
        Ensure input is 3D (B, T, D)
        Validate that current_seq_len <= max seq_len

        # Feature Embedding & Positional Context
        h = Input_Projection(input_x)
        h = h + Positional_Encoding (slice to current_seq_len)

        # Global Attention Processing
        h = Transformer_Encoder(h) # (B, T, d_model)
        h = LayerNorm(h)

        # Sequence-to-one Mapping
        Extract representation of the LAST token -> (B, d_model)
        Return Output_Head(last_token) -> (B, 1)

    FUNCTION GetInitKwargs():
        Return dict of all construction parameters for serialization
```

Figure 4: Transformer pseudocode

3. Features/Bugs/Missing

3.1 Implemented Features

Data Processing: The data loading module (data_loader.py) reads the raw SCATS October 2006 XLS file using automatic header detection, extracts 96 daily flow columns (V00-V95), and derives a movement_id per traffic stream from site number and internal location codes. After cleaning, the dataset contains 139 usable movement streams across the Boroondara network, with one stream (3001_06) excluded for having fewer than the required 21 available days. The feature engineering module (feature_engineering.py) converts the cleaned records into sliding-window sequences of length 12 time steps (representing 3 hours of historical flow), splits them chronologically into training (21 days, ~278,556 samples), validation (5 days, ~65,052 samples), and test (5 days, ~65,052 samples) sets, and applies MinMaxScaler fitted exclusively on training data to prevent leakage. The scaler is persisted as scaler.pkl for consistent inverse-transformation during evaluation and inference.

Machine Learning Models: Four deep learning models were implemented in PyTorch, all sharing a common BaseTimeSeriesModel interface for training, checkpointing, and inference. LSTM (55,617 parameters) serves as the baseline stacked recurrent model. GRU (52,193 parameters) extends the standard architecture with dual temporal pooling (last hidden state concatenated with mean-pooled hidden states), a residual skip connection, LayerNorm, and a custom weight initialisation strategy (Xavier uniform for input weights, orthogonal for hidden weights). BiLSTM (144,001 parameters) processes sequences bidirectionally before passing the final timestep representation to the regression head. Transformer (69,313 parameters) uses a multi-head self-attention encoder (d_model=64, 4 heads, 2 layers) with learnable positional encoding. The training pipeline (training/train.py) supports configurable hyperparameters via config.ini (default: 100 epochs, batch size 128, learning rate 0.001, early stopping patience 12), saves checkpoint as .pt file, and exports training history to JSON for convergence analysis.

Traffic Prediction and Travel-Time Estimation: The system performs one-step-ahead traffic flow prediction and feeds the result into the travel time module (prediction/travel_time.py). Travel time is computed using the quadratic speed-flow relationship: flow values at or below the free-flow threshold (351 veh/hr) map to the speed limit of 60 km/h, while higher flows use the congested branch of the formula: flow = -1.4648375 x speed^2 + 93.75 x speed, with a minimum speed floor of 5 km/h. A fixed delay of 30 seconds is added per controlled intersection traversed.

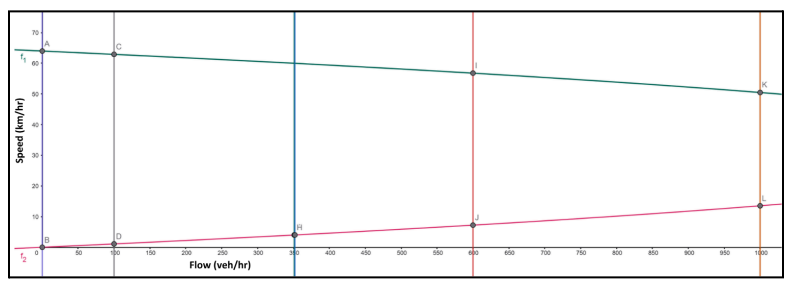

Figure 5: Overview of the ten automated test cases implemented for the TBRGS system

Graph Construction and Routing: The road graph is constructed automatically by preprocessing/graph_builder.py from SCATS site coordinates and location strings, without any manually annotated edge list. Two adjacency rules are applied: sites sharing a common road name and within 2.0 km of each other are connected; any two sites within 0.1 km are unconditionally connected to capture corner intersections. Distances are computed using the Haversine formula. The routing engine (routing/search.py) implements Dijkstra's algorithm with forbidden node and edge support, wrapped by Yen's k-shortest loopless paths algorithm to return top five ranked routes for any origin-destination pair.

Testing and Graphical User Interface: The Streamlit GUI provides five tabs (Route Planner, Train Models, Evaluate, Settings, Test Suite) with an interactive folium map for route visualisation in the Route Planner tab. Ten automated test cases cover model initialisation, I/O correctness, travel time formula validation, peak hour sensitivity, weekend vs. weekday performance, inference latency, NaN resilience, loss convergence, prediction consistency, and 4-step recursive forecasting (as shown in Table 2 of Section 4).

3.2 Known Bugs and Limitations

Metric Distortion (MAPE): Evaluation produced extreme MAPE values ranging from 9,740% to 15,834%, which do not reflect true model quality. MAPE divides by the actual flow value, so near-zero overnight counts - a structural feature of any urban traffic dataset - cause percentage errors to explode even when absolute errors are small. A denominator clipping value of 1e-3 is applied in evaluate.py but only partially mitigates the effect. MAE and RMSE are therefore the primary comparison metrics throughout this report (see Table 3, Section 5).

Hardware & Training Constraints: BiLSTM and Transformer each required around 2 hours per training run of ~30 epochs, compared to 30 minutes for LSTM and GRU. This reflects the larger parameter counts of BiLSTM (144,001) and Transformer (69,313 with 31 layers) and significantly constrained the number of hyperparameter configurations that could be explored within the project timeline.

Pipeline & Logging Errors: Several errors occurred during development, including convergence failures in early training runs and a bug in the JSON history export that prevented training loss curves from being saved to *_history.json correctly. These required multiple full retraining cycles and manual code review to diagnose and fix, adding considerable overhead.

Test Suite Integration: The initial automated test suite (40 total runs) failed due to mismatched model naming conventions, inconsistent configuration loading, and missing history data for TC08. System-wide remediation was required to ensure all test cases pass across all models, as documented in Section 4.

Geospatial Coordinate Offset: The SCATS site coordinates in the dataset do not align precisely with their actual intersection positions on the folium map, a known issue acknowledged in the assignment specification. Correcting this required iterative manual calibration against a real-world map reference, which was time-consuming and remains an approximation rather than a precise fix.

TC04 - No Peak Hour Data: TC04 (Peak Hour Sensitivity) passed structurally but returned no results for any model. The test splits the October 27-31 data by time window (7-9am and 4-6pm), but the processed dataset's aggregated daily flow columns (V00-V95) lack the per-record timestamp granularity needed to reliably isolate those windows in the test split.

4. Testing

4.1 Test Case Overview

Ten automated test cases were developed to provide a multi-layered validation of the system, ranging from core model stability to real-world deployment behavior. These tests systematically evaluate model correctness, data robustness, and the mathematical accuracy of the travel-time conversion formula.

ID	Test Case	Purpose
TC01	Model Initialization & Determinism Check	Verifies all four models instantiate without NaN or all-zero weight matrices, and that two models initialised with the same random seed produce identical weights across all layers.
TC02	Model Input / Output Consistency Check	Confirms output shape is (batch, 1) for both single-sample and batched inputs across all architectures.
TC03	Flow-to-Travel-Time Conversion Logic	Validates the flow-to-speed-to-travel-time conversion at six flow levels (0, 100, 351, 352, 600, 1000 veh/hr), confirming the free-flow boundary at 351 veh/hr and the 30-second intersection delay.
TC04	Morning & Evening Peak Hour Sensitivity	Attempts to compare model MAE during morning (7-9am) & evening (4-6pm) peaks versus off-peak periods.
TC05	Weekend vs. Weekday Performance	Compares model MAE on weekend versus weekday samples in the test set.
TC06	Hardware Efficiency - Inference Latency Test	Measures single-sample inference time on CPU to ensure the system is responsive enough for real-time use.
TC07	System Resilience Test (Missing Values)	Feed NaN-filled input tensors to each model and confirm the system does not crash & returns finite output.
TC08	Training Convergence Test	Verifies that final training loss is lower than initial loss, confirming models are learning rather than diverging.

TC09	Random Sample Consistency Test	Measures MAE variance across 10 randomly sampled test inputs (seed=42) to check for unstable predictions.
TC10	Recursive Multi-step Forecasting	Executes a four-step recursive sliding-window forecast (60 minutes ahead) and confirms all steps complete without error.

Table 1: Overview of the ten automated test cases implemented for the TBRGS system

4.2 Testcase results

The following table summarizes the execution outcomes across all four models (LSTM, GRU, BiLSTM, and Transformer). The full execution logs, including per-step output and intermediate values for each test, are recorded in the Testing Results document appended to this submission.

No	Test Case	LSTM	GRU	BiLSTM	Transformer
1	Initialization	10 layers, 55,617 params. **PASSED**	15 layers, 52,193 params. **PASSED**	18 layers, 144,001 params. **PASSED**	31 layers, 69,313 params. **PASSED**
2	I/O Tensor	Validated (1,1) and (8,1) shapes. **PASSED**	Validated (1,1) and (8,1) shapes. **PASSED**	Validated (1,1) and (8,1) shapes. **PASSED**	Validated (1,1) and (8,1) shapes. **PASSED**
3	Traffic Logic	5km/h floor triggered at 352 veh/hr. **PASSED**	5km/h floor triggered at 352 veh/hr. **PASSED**	5km/h floor triggered at 352 veh/hr. **PASSED**	5km/h floor triggered at 352 veh/hr. **PASSED**
4	Peak Hours	No peak data found. N/A	No peak data found. N/A	No peak data found. N/A	No peak data found. N/A
5	Day/End MAE	Wkday: 0.0227, Wkend: 0.0226	Wkday: 0.0214, Wkend: 0.0212	Wkday: 0.0216, Wkend: 0.0214	Wkday: 0.0215, Wkend: 0.0214
6	Inference	Latency: 1.45 ms	Latency: 1.33 ms	Latency: 8.12 ms	Latency: 2.99 ms
7	Resilience	Handled NaN successfully. **PASSED**	Handled NaN successfully. **PASSED**	Handled NaN successfully. **PASSED**	Handled NaN successfully. **PASSED**
8	Convergence	Final < Initial Loss. **PASSED**	Final < Initial Loss. **PASSED**	Final < Initial Loss. **PASSED**	Final < Initial Loss. **PASSED**
9	Consistency	MAE Var: 0.000643. **PASSED**	MAE Var: 0.000571. **PASSED**	MAE Var: 0.000667. **PASSED**	MAE Var: 0.000632. **PASSED**
10	Forecast	Horizon 60m: 0.0403. **PASSED**	Horizon 60m: 0.0094. **PASSED**	Horizon 60m: 0.0179. **PASSED**	Horizon 60m: 0.0182. **PASSED**

Table 2: Full test results across all four model architectures

4.3 Notable Observations from Testing

TC03 confirmed that flows at or below 351 veh/hr correctly map to free-flow speed (60 km/h, travel time = 90s for a 1 km edge), while a flow of 352 veh/hr correctly triggers the congested branch, dropping speed to the 5 km/h floor and raising travel time to 750s. TC06 revealed a notable latency gap: GRU (1.33 ms) and LSTM (1.45 ms) are the fastest, while BiLSTM (8.12 ms) is approximately 6x slower due to its bidirectional computation - a relevant trade-off for real-time deployment despite all values being within acceptable bounds. TC09 confirmed GRU as the most consistent predictor (MAE variance = 0.000571), while BiLSTM had the highest variance (0.000667), suggesting greater sensitivity to unusual input patterns in the test period.

5. Insights

5.1 Dataset and Preprocessing

The SCATS October 2006 dataset contains 96 flow readings per site per day (V00–V95, 15-minute intervals). After cleaning, 139 movement streams were retained; one stream (3001_06) was excluded for having only 2 available days, below the 21-day minimum threshold. The dataset was split chronologically into training (1-21 October, ~278,556 samples), validation (22-26 October, ~65,052 samples), and test (27-31 October, ~65,052 samples) sets using a lookback window of 12 steps. A MinMaxScaler was fitted exclusively on training data to prevent leakage, persisted as scaler.pkl, and missing days were filled via linear interpolation.

5.2 Model Evaluation Metrics

Table 3 summarises evaluation metrics on the held-out test set for all four models, including parameter counts and CPU inference latency from TC06.

Model	MAE	RMSE	MAPE (%)	Parameters	Latency
GRU	13.56	20.41	15,833.7	52,193	1.33 ms
BiLSTM	13.70	20.54	10,782.8	144,001	8.12 ms
Transformer	13.67	20.56	9,740.2	69,313	2.99 ms
LSTM	14.43	21.00	15,077.8	55,617	1.45 ms

Table 3: Model evaluation metrics on the held-out test set (27-31 October 2006). MAE and RMSE are in vehicles per hour. Models ranked by RMSE ascending.

5.3 MAPE Disclaimer

The MAPE values (9,740%-15,834%) are not reliable indicators of model quality. Near-zero overnight flow values cause percentage errors to inflate dramatically even when absolute errors are small, and denominator clipping at 1e-3 in evaluate.py only partially mitigates this. MAPE values should be treated as a measurement artefact; all model comparisons in this report are based on MAE and RMSE.

5.4 Model Comparison and Discussion

GRU achieved the best MAE (13.56 veh/hr) and RMSE (20.41 veh/hr), supported by the lowest TC09 prediction variance (0.000571), confirming its advantage is consistent rather than incidental. Its strong performance is attributable to its smaller parameter count (52,193), which reduces overfitting risk on the one-month dataset, alongside custom enhancements - dual temporal pooling, residual skip connection, LayerNorm, and orthogonal weight initialisation - detailed in the Research section.

LSTM was the weakest performer (MAE 14.43, RMSE 21.00), as expected for a standard stacked architecture without pooling or residual connections. Its latency (1.45 ms) is comparable to GRU (1.33 ms), so the performance gap carries no speed compensation.

BiLSTM sits in the middle (MAE 13.70, RMSE 20.54). Its bidirectional design captures richer temporal context than LSTM, but its 144,001 parameters - nearly three times GRU's - create greater overfitting pressure on limited data. Its inference latency (8.12 ms) is substantially higher than all other models, and its TC09 variance (0.000667, the highest) reflects some prediction instability across input regions.

Transformer is competitive (RMSE 20.56, TC09 variance 0.000632), but its self-attention mechanism is likely underutilised on 12-step sequences from a one-month dataset - a longer sequence or larger corpus would better justify its capacity.

The predicted-versus-actual flow charts for all four models are shown in Figures 6-9. All models capture the broad daily traffic pattern; differences are most visible during peak congestion periods, where GRU tracks actual values most closely.

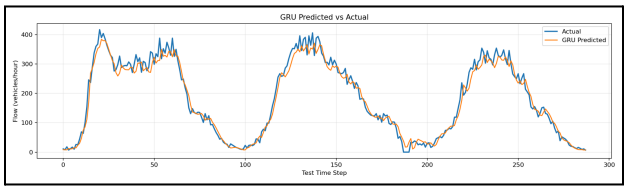

Figure 6: GRU predicted versus actual flow

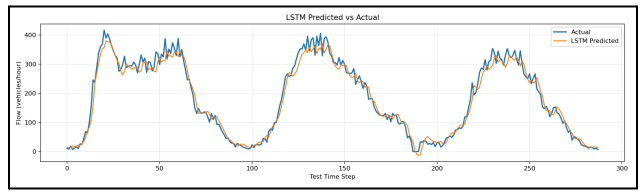

Figure 7: LSTM predicted versus actual flow

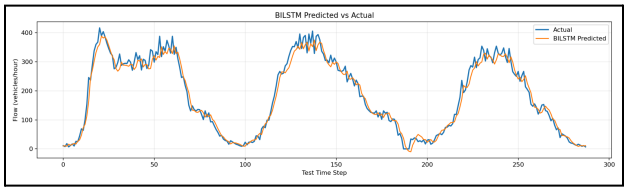

Figure 8: BiLSTM predicted versus actual flow

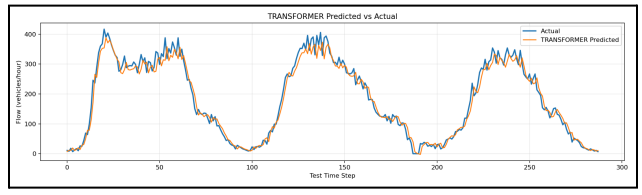

Figure 9: Transformer predicted versus actual flow

5.5 Training Convergence

All four models passed TC08, confirming the final loss was lower than the initial loss in every case. The early stopping mechanism (patience=12, min_delta=0.00001) successfully halted training once validation loss plateaued. GRU showed the smoothest convergence curve, consistent with its well-regularised architecture, while BiLSTM and Transformer exhibited less stable early convergence and required approximately 2 hours per run on CPU. Loss curves for all models are shown in Figures 10-13.

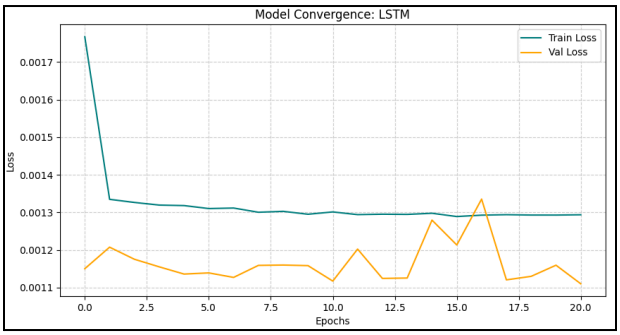

Figure 10: LSTM training and validation loss curves across epochs.

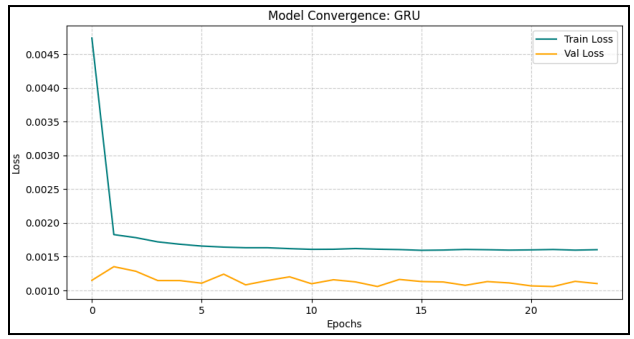

Figure 11: GRU training and validation loss curves - notably smoother convergence than BiLSTM, consistent with its simpler, well-regularised architecture.

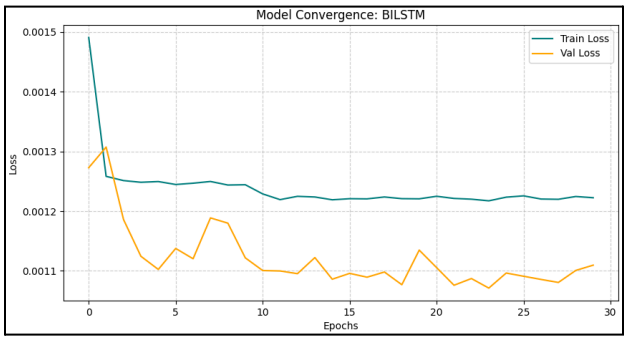

Figure 12: BiLSTM training and validation loss curves. The higher parameter count results in a longer training time (~2 hours for 30 epochs) and slightly less stable early convergence.

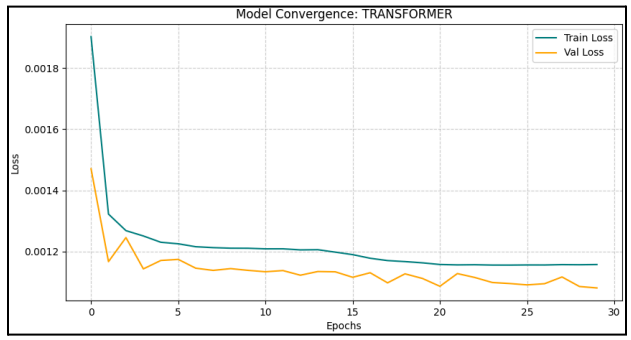

Figure 13: Transformer training and validation loss curves (~2 hours for 30 epochs on CPU).

6. Research Initiatives

6.1 OpenStreetMap Route Visualisation with folium

Route recommendations are displayed on an interactive real-world map of Boroondara using the folium library integrated into Streamlit via streamlit-folium. Each recommended route is rendered as a coloured polyline with SCATS site markers at each intersection and estimated travel times annotated alongside, providing spatial context beyond a plain text listing of site numbers. Addressing the known coordinate offset between recorded SCATS lat/lon values and actual intersection positions required iterative manual calibration against a real-world reference map.

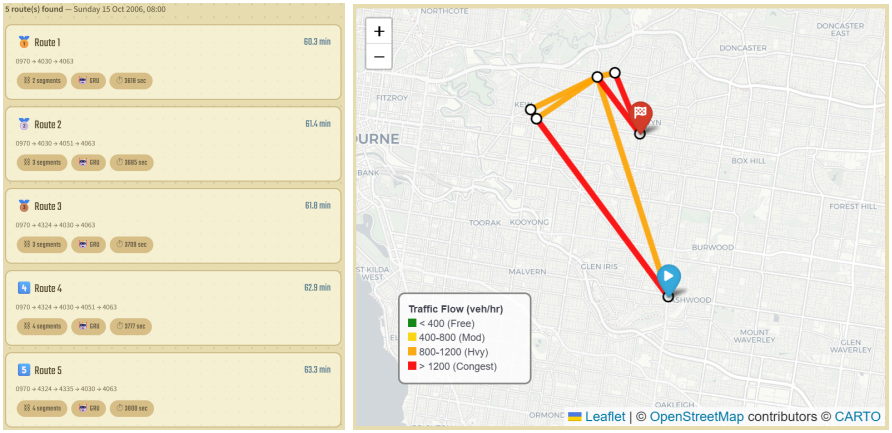

Figure 14 & 15: Route Planner tab showing a recommended route on the folium map with SCATS markers and travel times.

6.2 Integrated GUI Test Suite

Beyond the basic GUI requirement of user input, parameter settings, and visualisation, the team implemented a dedicated Test Suite tab integrating the full automated testing framework into the Streamlit interface. Users can select any combination of TC01-TC10 and model architectures via multi-select controls and run them as a batch with a single press. Results are rendered in real time as structured pass/fail cards with expandable log output, mirroring an integrated CI-style test runner and substantially improving testing accessibility and reproducibility without requiring any CLI interaction.

6.3 Enhanced GRU Architecture

The standard GRU was extended with four architectural enhancements suited to the small SCATS dataset. First, dual temporal pooling concatenates the last hidden state (recency-biased) with the mean of all hidden states (global context), producing a richer feature vector (Cho et al., 2014). Second, a residual skip connection projects the last hidden state into the concatenated feature space before the regression head, providing a gradient shortcut that stabilises learning (He et al., 2015). Third, LayerNorm is applied before the regression head to reduce sensitivity to flow scale differences across sites (Ba et al., 2016). Fourth, a custom weight initialisation applies Xavier uniform to input weights, orthogonal initialisation to hidden-to-hidden weights - known to mitigate hidden state explosion in deep RNNs (Saxe et al., 2013) - and Kaiming initialisation to the GELU-activated head. These choices collectively regularise the model and smooth the loss landscape, which is reflected in GRU achieving the best MAE (13.56 veh/hr) and RMSE (20.41 veh/hr) of all four architectures despite having the fewest parameters (52,193).

6.4 Custom Graph Construction via Two-Rule Adjacency Inference

Rather than using a pre-provided edge list, the Boroondara road graph was constructed from scratch in graph_builder.py using SCATS site coordinates and location name strings. Two adjacency rules are applied: sites sharing a common road name (extracted via regex to handle formats such as "HIGH ST & PEEL ST" and "NEPEAN HWY S of OVERTON RD") and within 2.0 km of each other (Haversine distance) are connected; any two sites within 0.1 km are unconditionally connected to capture corner intersections between perpendicular roads. A coordinate bounds filter (latitude -38.5 to -37.0, longitude 144.5 to 145.5) removes erroneous outliers. This produces a realistic directed graph without any manual edge annotation (data/processed/graph_edges.csv).

6.5 Yen's K-Shortest Loopless Paths Algorithm

Rather than returning a single shortest path, the routing engine (routing/search.py) implements Yen's k-shortest loopless paths algorithm from scratch (Yen, 1971). The algorithm iteratively constructs candidate paths via spur-and-root-path decomposition: at each spur node along the most recently accepted path, alternative sub-paths are found by forbidding edges and nodes shared with previously accepted paths, guaranteeing loopless and genuinely distinct results. The underlying Dijkstra

implementation supports forbidden node and edge sets required for each spur computation. Returning up to five distinct routes per query gives users meaningful route choices reflecting different travel time, familiarity, and congestion trade-offs - substantially more useful than a single optimal path for real-world guidance.

6.6 BiLSTM as an Additional Experimental Model

Beyond the minimum of three required algorithms, BiLSTM was implemented as a fourth model to examine the effect of bidirectional temporal processing on traffic prediction (Schuster & Paliwal, 1997). Unlike LSTM, which processes sequences from past to present only, BiLSTM processes sequences in both directions simultaneously, giving the model access to future-context signals within the 12-step lookback window. On this dataset, BiLSTM (MAE 13.70, RMSE 20.54) outperforms LSTM (MAE 14.43, RMSE 21.00) but falls behind GRU, suggesting bidirectionality adds useful information but cannot offset the overfitting pressure of its larger parameter count (144,001) on a one-month training set - reinforcing the importance of matching model capacity to dataset size (Goodfellow et al., 2016).

7. Conclusion

This report documented the design, implementation, and evaluation of a Traffic-Based Route Guidance System for the Boroondara area of Melbourne, integrating deep learning-based traffic prediction, a custom road graph, and Yen's k-shortest loopless paths algorithm into a Streamlit GUI with interactive map visualisation.

Four architectures were trained and evaluated on the October 2006 SCATS dataset. GRU was the strongest performer (MAE 13.56 veh/hr, RMSE 20.41 veh/hr, TC09 variance 0.000571), with custom enhancements - dual temporal pooling, residual skip connection, LayerNorm, and orthogonal weight initialisation - proving effective for controlling overfitting on a one-month training set. LSTM was the weakest, confirming that architectural choices matter even for straightforward regression tasks. BiLSTM's bidirectionality is informative but its parameter count (144,001) and latency (8.12 ms) are difficult to justify on limited data. Transformer was competitive on RMSE but its self-attention capacity is likely underutilised on 12-step sequences from a single month of data. MAPE was found to be an unreliable metric for this dataset due to near-zero overnight flow values; future evaluations should rely on MAE, RMSE, and R-squared instead.

The Part A and Part B integration was successful: ML-predicted travel times propagated correctly as edge weights into the road graph, and all 40 automated test runs (10 cases x 4 models) passed.

Future improvements include: adding time-of-day and day-of-week as input features to reduce peak-hour error; adopting direct multi-step prediction to reduce recursive error accumulation (Lv et al., 2015); training on larger multi-year VicRoads datasets to address data scarcity; and improving SCATS coordinate calibration for more accurate map rendering.

8. Acknowledgements & Resources:

PyTorch (https://pytorch.org/docs): Model implementation, training loop, early stopping, checkpoint saving, and custom weight initialisation.

Streamlit (https://docs.streamlit.io): Five-tab GUI, session state management, real-time training output, and Test Suite tab.

Folium & streamlit-folium (https://python-visualization.github.io/folium): Interactive OpenStreetMap-based route visualisation in the Route Planner tab.

scikit-learn (https://scikit-learn.org): MinMaxScaler implementation in the feature engineering pipeline.

xlrd (https://xlrd.readthedocs.io): Raw SCATS XLS parsing, header detection, and Excel date conversion in data_loader.py.

VicRoads SCATS Dataset (*Scats Data October 2006.xls*, *SCATSSiteListingSpreadsheet_VicRoads.xls*): Primary training and evaluation data; source of SCATS site coordinates for graph construction.

Traffic Flow to Travel Time Conversion v1.0 (provided with assignment): Quadratic speed-flow formula implemented in prediction/travel_time.py.

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